

Notebook

EBN III

NOTES



Study Planner

Goal : 65% Actual :

study goals

- 1 TESTS EACH WEEK
- 2 PRACTICE PROBLEMS
- 3 PRACTICE Assignments

to do list

ASSIGNMENT #4
WORK CATCH UP

reminder

time

Schedule

notes

CH. 3 $\frac{1}{3}$ CH. 10

NB! 

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Topic or Essential Question: **CHAPTER 3.6 :**
NODAL ANALYSIS VIA INSPECTION

Questions and Ideas	Notes:
WHY IS G_{AB} NEGATIVE?	NODAL ANALYSIS VIA INSPECTION USES THE CONDUCTANCE OF EACH RESISTOR THE CONDUCTANCE @ A NODE = $\sum G$ CONNECTED TO THE NODE $G_A = G_1 + G_2 \quad \text{①}$ $G_B = G_2 + G_3 \quad \text{②}$ $G_{AB} = -G_2 \quad \text{③}$ <u>CONDUCTANCE MATRIX</u> $\begin{matrix} \text{①} & \text{②} & \text{VOLTAGE @ NODE} \\ \begin{matrix} \text{①} \\ \text{②} \end{matrix} & \begin{bmatrix} G_A & G_{AB} \\ G_{BA} & G_B \end{bmatrix} & \begin{bmatrix} V_A \\ V_B \end{bmatrix} = \begin{bmatrix} I_1 - I_2 \\ I_2 \end{bmatrix} \end{matrix}$ CURRENT ENTER $\oplus$ CURRENT LEAVE $\ominus$ CURRENT @ NODES $\rightarrow G_A V_A + G_{AB} V_B = I_1 - I_2$ $\rightarrow G_{BA} V_A + G_B V_B = I_2$

Summary: NODAL ANALYSIS VIA INSPECTION USES THE CONDUCTANCE OF EACH RESISTOR. IT TYPICALLY HAS CURRENT SOURCES.

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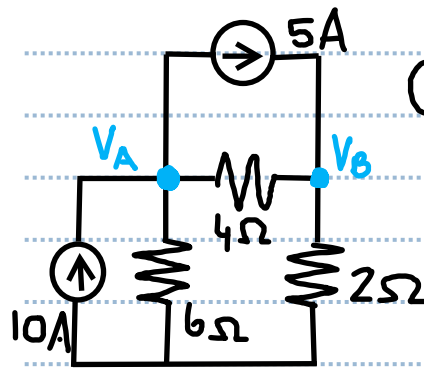
Topic or Essential Question: _____

Questions and Ideas

Notes:

@Example 1.1;
Equation 3:
Why is it $-\frac{1}{4}$
and not $+\frac{1}{4}$

EXAMPLE 1.1:



$$G_A = \frac{1}{6} + \frac{1}{4} = \frac{5}{12} \quad \text{①}$$

$$G_B = \frac{1}{4} + \frac{1}{2} = \frac{3}{4} \quad \text{②}$$

$$G_{AB} = -\frac{1}{4} \quad \text{③}$$

* WITH THIS METHOD
WE TYPICALLY HAVE
CURRENT SOURCES

← Nodal

@ Conductance
Matrix, Why are
both G_{AB} and G_{BA}
0? Are G_{AB} & G_{BA}
not in opposite
directions?

$$\begin{bmatrix} \frac{5}{12} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{3}{4} \end{bmatrix} \begin{bmatrix} V_A \\ V_B \end{bmatrix} = \begin{bmatrix} 10-5 \\ 5 \end{bmatrix}$$

$$\rightarrow D \left[\frac{5}{12} \times \frac{3}{4} \right] - \left[-\frac{1}{4} - (-\frac{1}{4}) \right]$$

$$D = \frac{1}{4}$$

$$\rightarrow G_A = \left[5 \times \frac{3}{4} \right] - \left[-\frac{1}{4} \times 5 \right]$$

$$G_A = 5$$

$$\rightarrow G_B = \left[\frac{5}{12} \times 5 \right] - \left[5 \times -\frac{1}{4} \right]$$

$$= 3,33$$

$$V_A = \frac{G_A}{D} = 20V$$

$$V_B = \frac{G_B}{D} = 13,33V$$

Summary: _____

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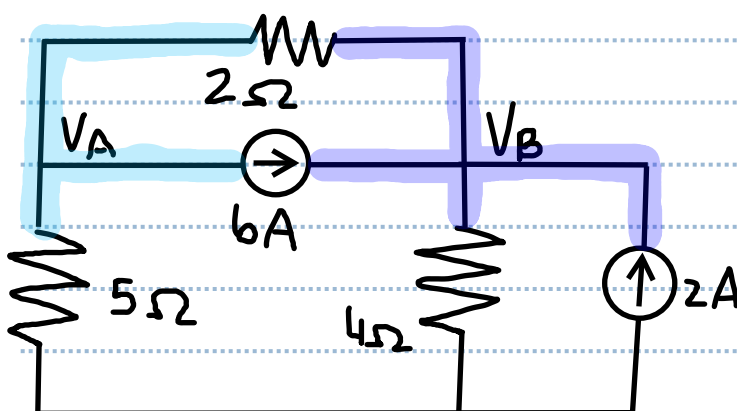
Topic or Essential Question:

Questions and Ideas

Notes:

SHOULD YOU ASSUME
ALL CURRENTS ARE
FLOWING OUT (LIKE
IN NORMAL NODAL)

EXAMPLE #2:



$G_A =$

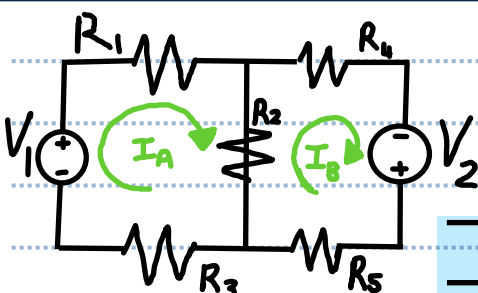
Summary:

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Topic or Essential Question: CHAPTER 3

MESH ANALYSIS VIA INSPECTION

Questions and Ideas	Notes:
	 FIND RESISTANCE MATRIX ELEMENTS: → $R_A = \sum R \text{ IN MESH A}$ → $R_B = \sum R \text{ IN MESH B}$ → $R_{AB} = R \text{ SHARED BY BOTH MESSES}$ $\therefore R_A = R_1 + R_2 + R_3$ $R_B = R_2 + R_4 + R_5$ $R_{AB} = -R_2$ $\begin{bmatrix} R_A & R_{AB} \\ R_{BA} & R_B \end{bmatrix} \begin{bmatrix} I_A \\ I_B \end{bmatrix} = \begin{bmatrix} V_1 \\ -V_2 \end{bmatrix}$

Summary: MESH ANALYSIS VIA INSPECTION TYPICALLY HAS VOLTAGE SOURCES

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Topic or Essential Question:

[illegible]

Summary:

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Topic or Essential Question:

[illegible]

Summary:

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Topic or Essential Question:

[illegible]

Summary:

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Topic or Essential Question:

[illegible]

Summary:

Cornell Note-Taking



Topic or Essential Question:

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Summary:

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Topic or Essential Question:

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Topic or Essential Question:

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Topic or Essential Question:

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